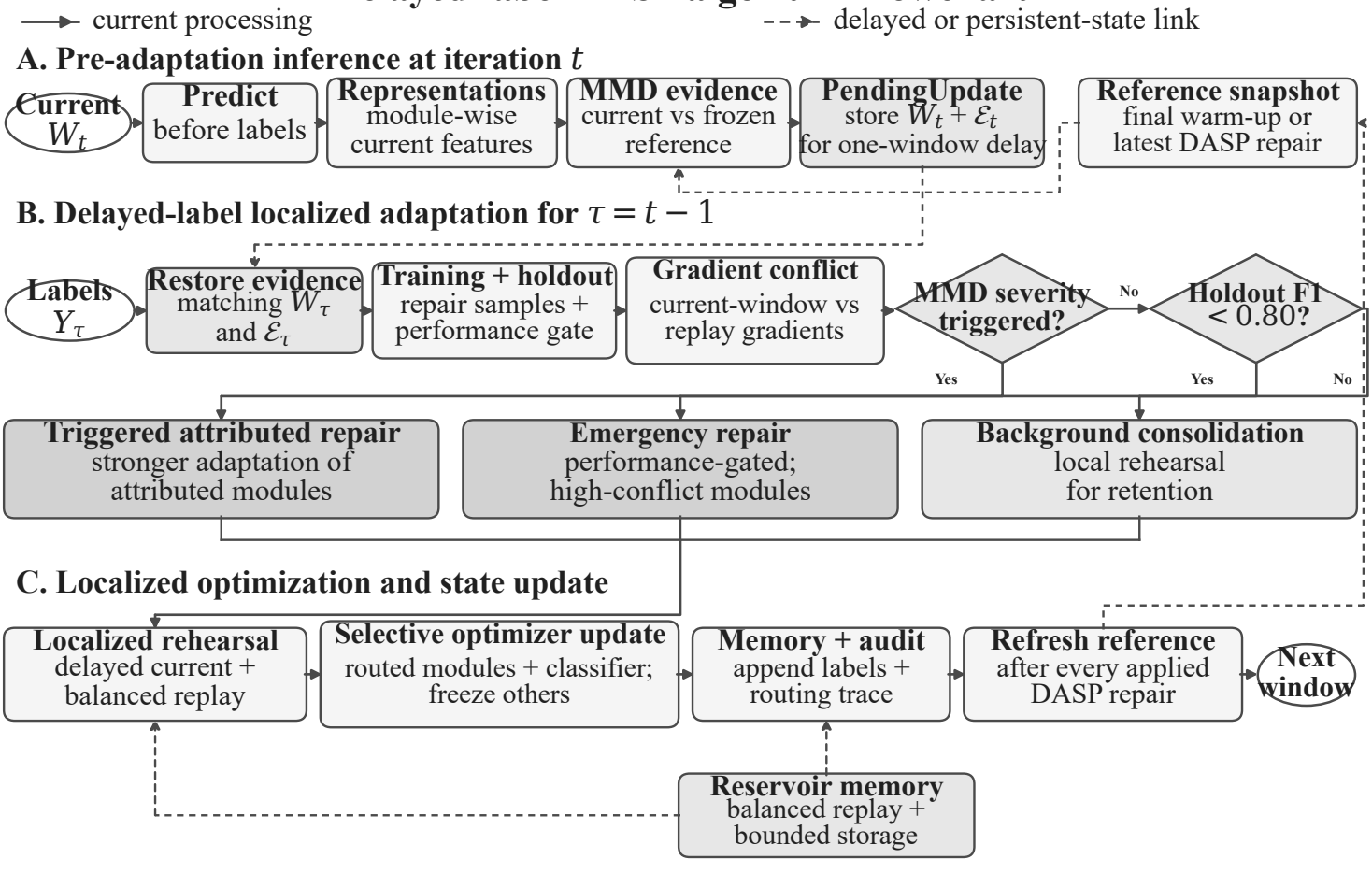

Delayed-label DASP algorithm flowchart
current processing
delayed or persistent-state link
A. Pre-adaptation inference at iteration t
Current W_t
Predict
before labels
Representations
module-wise
current features
MMD evidence
current vs frozen
reference
PendingUpdate
store $W_t + \mathcal{E}_t$
for one-window delay
Reference snapshot
final warm-up or
latest DASP repair
B. Delayed-label localized adaptation for $\tau = t - 1$
Labels Y_τ
Restore evidence
matching W_τ
and $\mathcal{E}_\tau$
Training + holdout
repair samples +
performance gate
Gradient conflict
current-window vs
replay gradients
MMD severity
triggered?
No
Holdout F1
< 0.80?
Yes
Yes
No
Triggered attributed repair
stronger adaptation of
attributed modules
Emergency repair
performance-gated;
high-conflict modules
Background consolidation
local rehearsal
for retention
C. Localized optimization and state update
Localized rehearsal
delayed current +
balanced replay
Selective optimizer update
routed modules + classifier;
freeze others
Memory + audit
append labels +
routing trace
Refresh reference
after every applied
DASP repair
Next
window
Reservoir memory
balanced replay +
bounded storage